1. The permutation of *n* objects *k* at a time can be defined as shown below. It tells us how many permutations of *n* objects can be formed *k* elements at a time.

$$P(n, k) = \text{factorial}(n) / \text{factorial}(n - k)$$

Output:

Enter a positive integer for n (1 to 20): 6

Enter a positive integer for k (k <= n): 4

Permutation of n object k at a time: 360

2. The Pascal triangle defines the coefficient of the terms (C_n) in the expansion of the binomial

$$(x + y)^n = C_0\, x^0 y^n + C_1\, x^1 y^{n-1} + \ \ldots\ + C_{n-1}\, x^{n-1} y^1 + C_n\, x^n y^0$$

The following shows the coefficient related to each *n* from 0 to 5. Note that both rows and columns starts from 0.

n = 0	1					
n = 1	1	1				
n = 2	1	2	1			
n = 3	1	3	3	1		
n = 4	1	4	6	4	1	
n = 5	1	5	10	10	5	1

Note that the value of each coefficient in a row and column is the sum of the coefficients in the previous row and previous column added with the coefficient in the previous row and the same column as shown below.

$$\text{Pascal}(row, col) = \text{pascal}(row - 1, col - 1) + \text{pascal}(row - 1, col)$$

Write a function to find the value's coefficients for any *n*. Test your function to print the coefficient for *n* from 1 to 10 in rows and columns.

2

Output:

Enter the size of the Pascal triangle: 5

The coefficients are shown below:

```
1
1   1
1   2   1
1   3   3   1
1   4   6   4   1
1   5   10  10  5   1
```

3. An integer is *emirp* (prime spelled backward) if when we inverse the digits, it is also a prime number. Write a function to print the *emirp* integers from 1 to 1000 ten numbers in a line.

Output:

The emirp numbers less than 1000:

2 3 5 7 11 13 17 31 37 71
73 79 97 101 107 113 131 149 151 157
167 179 181 191 199 311 313 337 347 353
359 373 383 389 701 709 727 733 739 743
751 757 761 769 787 797 907 919 929 937
941 953 967 971 983 991

4. Write a program that finds all the factors of a given number. A factor is a number no more than the given number that divides the number. The program should use a function to test if a number is divisible by another number. Test your function with numbers between 1 and 100.

Output:

Enter a number between 1 and 100: 60

The number: 60

The factors: 1 2 3 4 5 6 10 12 15 20 30 60